

Counting Tilings

Time limit: 1.00 s

Memory limit: 512 MB

Your task is to count the number of ways you can fill an $n \times m$ grid using 1×2 and 2×1 tiles.

Input

The only input line has two integers n and m .

Output

Print one integer: the number of ways modulo $10^9 + 7$.

Constraints

- $1 \leq n \leq 10$
- $1 \leq m \leq 1000$

Example

Input	Output
4 7	781